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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 16th March Evening Shift

JEE Main Question Paper with Answer Key

16 Mar 2021	90	20	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	23 questions
Mathematics	22 questions
Physics	25 questions
Out of syllabus	20 questions
Answer key	at the end

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Chemistry 23 questions

Q1. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II :

	List - I Test/Reagents/Observation(s)		List - II Species detected
(a)	Lassaigne's Test	(i)	Carbon
(b)	Cu(II) oxide	(ii)	Sulphur
(c)	Silver nitrate	(iii)	N, S, P, and halogen
(d)	The sodium fusion extract gives black precipitate with acetic acid and lead acetate	(iv)	Halogen Specifically

The correct match is :

(A) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

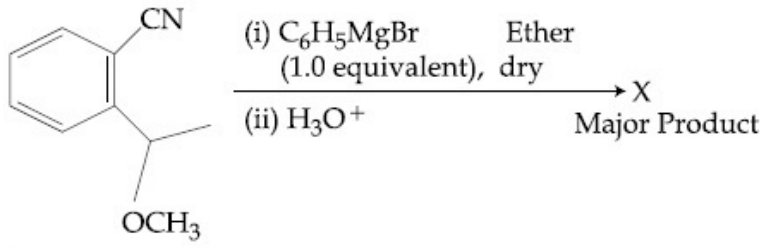
(B) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

(C) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(D) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

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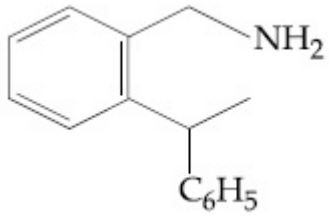
Q2. MCQ (Single Correct Answer)



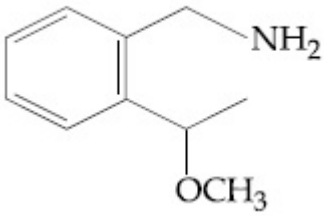
Open solution

The structure of X is :

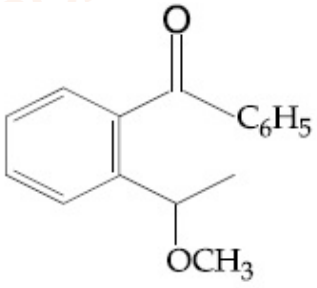
(A)



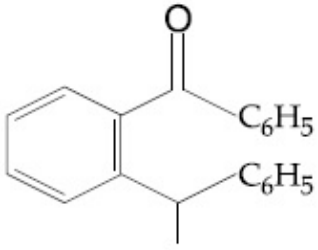
(B)



(C)



(D)



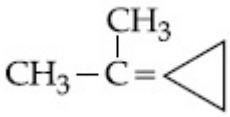
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Q3. MCQ (Single Correct Answer)

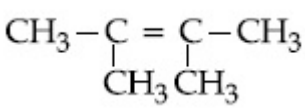
An unsaturated hydrocarbon X on ozonolysis gives A. Compound A when warmed with ammonical silver nitrate forms a bright silver mirror along the sides of the test tube. The unsaturated hydrocarbon X is :

Open solution

(A) $HC \equiv C - CH_2 - CH_3$

(B) 

(C) $CH_3 - C \equiv C - CH_3$

(D) 

Q4. MCQ (Single Correct Answer)

The exact volumes of 1 M NaOH solution required to neutralise 50 mL of 1 M H_3PO_3 solution and 100 mL of 2 M H_3PO_2 solution, respectively, are :

Open solution

(A) 100 mL and 50 mL

(B) 100 mL and 200 mL

(C) 100 mL and 100 mL

(D) 50 mL and 50 mL

Q5. MCQ (Single Correct Answer)

The characteristics of elements X, Y and Z with atomic numbers, respectively, 33, 53 and 83 are :

Open solution

(A) X and Y are metalloids and Z is a metal.

(B) X and Z are non-metals and Y is a metalloid.

(C) X is a metalloid, Y is a non-metal and Z is a metal.

(D) X, Y and Z are metals.

Q6. MCQ (Single Correct Answer)

Which of the following is least basic?

Open solution

(A) $(CH_3CO)_2\ddot{N}H$

(B) $(CH_3CO)\ddot{N}HC_2H_5$

(C) $(C_2H_5)_3\ddot{N}$

(D) $(C_2H_5)_2\ddot{N}H$

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Q7. MCQ (Single Correct Answer)

Ammonolysis of Alkyl halides followed by the treatment with NaOH solution can be used to prepare primary, secondary and tertiary amines. The purpose of NaOH in the reaction is :

(A) to remove acidic impurities

(B) to remove basic impurities

(C) to activate NH_3 used in the reaction

(D) to increase the reactivity of alkyl halide

Open solution

Q8. MCQ (Single Correct Answer)

The secondary structure of protein is stabilised by:

(A) Hydrogen bonding

(B) glycosidic bond

(C) Peptide bond

(D) van der Waals forces

Open solution

Q9. MCQ (Single Correct Answer)

Fex_2 and Fey_3 are known when x and y are :

(A) x = F, Cl, Br, I and y = F, Cl, Br, I

(B) x = F, Cl, Br and y = F, Cl, Br, I

(C) x = F, Cl, Br, I and y = F, Cl, Br

(D) x = Cl, Br, I and y = F, Cl, Br, I

Open solution

Q10. MCQ (Single Correct Answer)

Identify the elements X and Y using the ionisation energy values given below :

	Ionization energy (kJ/mol)	
	1st	2nd
X	495	4563
Y	731	1450

(A) X = F; Y = Mg

(B) X = Mg; Y = F

(C) X = Na; Y = Mg

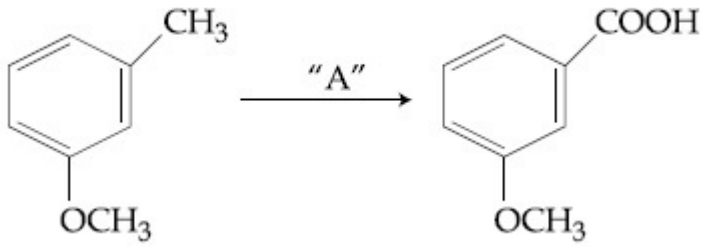
(D) X = Mg; Y = Na

Open solution

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Q11. MCQ (Single Correct Answer)

Open solution



In the above reaction, the reagent 'A' is :

(A) $LiAlH_4$

(B) $NaBH_4, H_3O^+$

(C) HCl, Zn – Hg

(D) Alkaline $KMnO_4, H^+$

Q12. MCQ (Single Correct Answer)

Open solution

Arrange the following metal complex/compounds in the increasing order of spin only magnetic moment. Presume all the three, high spin system.

(Atomic numbers Ce = 58, Gd = 64 and Eu = 63)

(a) $(NH_4)_2[Ce(NO_3)_6]$

(b) $Gd(NO_3)_3$ and

(c) $Eu(NO_3)_3$

(A) (a) < (b) < (c)

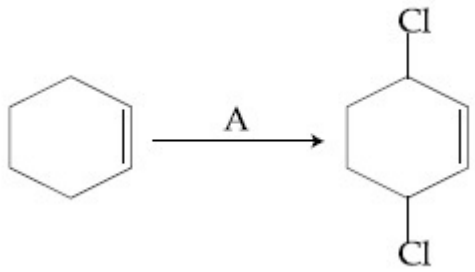
(B) (b) < (a) < (c)

(C) (a) < (c) < (b)

(D) (c) < (a) < (b)

Q13. MCQ (Single Correct Answer)

Open solution



Identify the reagent(s) 'A' and condition(s) for the reaction

(A) A = HCl, $ZnCl_2$

(B) A = HCl; Anhydrous $AlCl_3$

(C) A = Cl_2; UV light

(D) A = Cl_2; dark, Anhydrous $AlCl_3$

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Q14. MCQ (Single Correct Answer)

Statement I : Sodium hydride can be used as an oxidising agent.

Statement II : The lone pair of electrons on nitrogen in pyridine makes it basic.

Choose the CORRECT answer from the options given below :

(A) Statement I is false but statement II is true

(B) Both statement I and statement II are true

(C) Statement I is true but statement II is false

(D) Both statement I and statement II are false

Open solution

Q15. Numerical

$[Ti(H_2O)_6]^{3+}$ absorbs light of wavelength 498 nm during a d $-$ d transition. The octahedral splitting energy for the above complex is _____ $\times 10^{-19}$ J. (Round off to the Nearest Integer). h = 6.626×10^{-34} Js; c = 3×10^{8} ms^{-1}

Open solution

Q16. Numerical

A 5.0 m mol dm^{-3} aqueous solution of KCl has a conductance of 0.55 mS when measured in a cell of cell constant 1.3 cm^{-1}. The molar conductivity of this solution is _____ mSm^2 mol^{-1}. (Round off to the Nearest Integer).

Open solution

Q17. Numerical

At 25°C, 50 g of iron reacts with HCl to form $FeCl_2$. The evolved hydrogen gas expands against a constant pressure of 1 bar. The work done by the gas during this expansion is _____ J. (Round off to the Nearest Integer).

[Given : R = 8.314 J mol^{-1} K^{-1}. Assume, hydrogen is an ideal gas] [Atomic mass of Fe is 55.85 u]

Open solution

Q18. Numerical

The number of orbitals with n = 5, m_l = +2 is _____. (Round off to the Nearest Integer).

Open solution

Q19. Numerical

A and B decompose via first order kinetics with half-lives 54.0 min and 18.0 min respectively. Starting from an equimolar non reactive mixture of A and B, the time taken for the concentration of A to become 16 times that of B is _____ min. (Round off to the Nearest Integer).

Open solution

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Q20. Numerical

In Duma's method of estimation of nitrogen, 0.1840 g of an organic compound gave 30 mL of nitrogen collected at 287 K and 758 mm of Hg pressure. The percentage composition of nitrogen in the compound is _____. (Round off to the Nearest Integer). [Given : Aqueous tension at 287 K = 14 mm of Hg]

Open solution

Q21. Numerical

When 35 mL of 0.15 M lead nitrate solution is mixed with 20 mL of 0.12 M chromic sulphate solution, _____ $\times 10^{-5}$ moles of lead sulphate precipitate out. (Round off to the Nearest Integer).

Open solution

Q22. Numerical

Sulphurous acid (H_2SO_3) has $Ka_1 = 1.7 \times 10^{-2}$ and $Ka_2 = 6.4 \times 10^{-8}$. The pH of 0.588 M H_2SO_3 is _____. (Round off to the Nearest Integer).

Open solution

Q23. Numerical

At 363 K, the vapour pressure of A is 21 kPa and that of B is 18 kPa. One mole of A and 2 moles of B are mixed. Assuming that this solution is ideal, the vapour pressure of the mixture is _____ kPa. (Round off to the Nearest Integer).

Open solution

Mathematics 22 questions

Q24. MCQ (Single Correct Answer)

If y = y(x) is the solution of the differential equation

$\frac{dy}{dx}$ + (tan x) y = sin x, $0 \le x \le \frac{\pi}{3}$, with y(0) = 0, then $y\left(\frac{\pi}{4}\right)$ equal to :

(A) $\frac{1}{2}\log_e 2$

(B) $\left(\frac{1}{2\sqrt{2}}\right)\log_e 2$

(C) $\log_e 2$

(D) $\frac{1}{4}\log_e 2$

Open solution

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Q25. MCQ (Single Correct Answer)

Open solution

Let f be a real valued function, defined on R $-$ {$-$1, 1} and given by

f(x) = 3 $\log_e \left|\frac{x-1}{x+1}\right| - \frac{2}{x-1}$.

Then in which of the following intervals, function f(x) is increasing?

(A) $(-\infty, -1) \cup \left([\frac{1}{2}, \infty) - \{1\}\right)$

(B) $(-\infty, \infty) - \{-1, 1\}$

(C) $(-\infty, \frac{1}{2}] - \{-1\}$

(D) $(-1, \frac{1}{2}]$

Q26. MCQ (Single Correct Answer)

Open solution

Let $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} - 3\hat{j} + 5\hat{k}$. If $\vec{r} \times \vec{a} = \vec{b} \times \vec{r}$,

$\vec{r} \cdot \left(\alpha\hat{i} + 2\hat{j} + \hat{k}\right) = 3$ and $\vec{r} \cdot \left(2\hat{i} + 5\hat{j} - \alpha\hat{k}\right) = -1$, $\alpha \in$ R, then the

value of $\alpha + \left|\vec{r}\right|^2$ is equal to :

(A) 13 (B) 11 (C) 9 (D) 15

Q27. MCQ (Single Correct Answer)

Open solution

If the foot of the perpendicular from point (4, 3, 8) on the line $L_1 : \frac{x-a}{l} = \frac{y-2}{3} = \frac{z-b}{4}$, $l \neq 0$ is (3, 5, 7), then the shortest distance between the line L_1 and line $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is equal to :

(A) $\frac{1}{\sqrt{6}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{\frac{2}{3}}$

Q28. MCQ (Single Correct Answer)

Open solution

Consider a rectangle ABCD having 5, 7, 6, 9 points in the interior of the line segments AB, CD, BC, DA respectively. Let α be the number of triangles having these points from different sides as vertices and β be the number of quadrilaterals having these points from different sides as vertices. Then ($\beta - \alpha$) is equal to :

(A) 717 (B) 795 (C) 1890 (D) 1173

Q29. MCQ (Single Correct Answer)

Open solution

Let f : S $\to$ S where S = (0, ∞) be a twice differentiable function such that f(x + 1) = xf(x). If g : S $\to$ R be defined as g(x) = $\log_e$ f(x), then the value of |g"(5) $-$ g"(1)| is equal to :

(A) 1 (B) $\frac{187}{144}$ (C) $\frac{197}{144}$ (D) $\frac{205}{144}$

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Q30. MCQ (Single Correct Answer)

Consider the integral
$I = \int_0^{10} \frac{[x]e^{[x]}}{e^{x-1}}dx$,
where [x] denotes the greatest integer less than or equal to x. Then the value of I is equal to :

(A) 45 (e − 1) (B) 45 (e + 1) (C) 9 (e + 1) (D) 9 (e − 1)

Open solution

Q31. MCQ (Single Correct Answer)

Let C_1 be the curve obtained by the solution of differential equation

$2xy\frac{dy}{dx} = y^2 - x^2, x > 0$. Let the curve C_2 be the

solution of $\frac{2xy}{x^2-y^2} = \frac{dy}{dx}$. If both the curves pass through (1, 1), then the area enclosed by the curves C_1 and C_2 is equal to :

(A) $\frac{\pi}{4}$ + 1 (B) π + 1 (C) $\pi - 1$ (D) $\frac{\pi}{2} - 1$

Open solution

Q32. MCQ (Single Correct Answer)

Let P(x) = x^2 + bx + c be a quadratic polynomial with real coefficients such that $\int_0^1 P(x)dx$ = 1 and P(x) leaves remainder 5 when it is divided by (x − 2). Then the value of 9(b + c) is equal to :

(A) 9 (B) 11 (C) 7 (D) 15

Open solution

Q33. MCQ (Single Correct Answer)

Let A = {2, 3, 4, 5, ..., 30} and '$\simeq$' be an equivalence relation on A × A, defined by (a, b) $\simeq$ (c, d), if and only if ad = bc. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair (4, 3) is equal to :

(A) 5 (B) 6 (C) 8 (D) 7

Open solution

Q34. MCQ (Single Correct Answer)

The least value of |z| where z is complex number which satisfies the inequality
$\exp\left(\frac{(|z|+3)(|z|-1)}{||z|+1|}\log_e 2\right) \geq \log_{\sqrt{2}}|5\sqrt{7} + 9i|, i = \sqrt{-1}$, is equal to :

(A) 8 (B) 3 (C) 2 (D) $\sqrt{5}$

Open solution

Q35. MCQ (Single Correct Answer)

Let $\alpha \in$ R be such that the function $f(x) = \begin{cases} \frac{\cos^{-1}(1-\{x\}^2)\sin^{-1}(1-\{x\})}{\{x\}-\{x\}^3}, & x \neq 0 \\ \alpha, & x = 0 \end{cases}$ is continuous at x = 0,
where {x} = x − [x], [x] is the greatest integer less than or equal to x. Then :

(A) no such α exists (B) α = 0
(C) $\alpha = \frac{\pi}{4}$ (D) $\alpha = \frac{\pi}{\sqrt{2}}$

Open solution

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Q36. MCQ (Single Correct Answer)

Given that the inverse trigonometric functions take principal values only. Then, the number of real values of x which satisfy

$\sin^{-1}\left(\frac{3x}{5}\right)+\sin^{-1}\left(\frac{4x}{5}\right)=\sin^{-1}x$ is equal to :

Open solution

(A) 2 (B) 0 (C) 3 (D) 1

Q37. MCQ (Single Correct Answer)

Let the lengths of intercepts on x-axis and y-axis made by the circle $x^2 + y^2 + ax + 2ay + c = 0$, $(a < 0)$ be $2\sqrt{2}$ and $2\sqrt{5}$, respectively. Then the shortest distance from origin to a tangent to this circle which is perpendicular to the line $x + 2y = 0$, is equal to :

Open solution

(A) $\sqrt{10}$ (B) $\sqrt{6}$ (C) $\sqrt{11}$ (D) $\sqrt{7}$

Q38. MCQ (Single Correct Answer)

Let A$(-1, 1)$, B$(3, 4)$ and C$(2, 0)$ be given three points.
A line $y = mx$, $m > 0$, intersects lines AC and BC at point P and Q respectively. Let A_1 and A_2 be the areas of ΔABC and ΔPQC respectively, such that $A_1 = 3A_2$, then the value of m is equal to :

Open solution

(A) 1 (B) 3 (C) 2 (D) $\frac{4}{15}$

Q39. MCQ (Single Correct Answer)

Let A denote the event that a 6-digit integer formed by 0, 1, 2, 3, 4, 5, 6 without repetitions, be divisible by 3. Then probability of event A is equal to :

Open solution

(A) $\frac{4}{9}$ (B) $\frac{9}{56}$ (C) $\frac{11}{27}$ (D) $\frac{3}{7}$

Q40. MCQ (Single Correct Answer)

The maximum value of

$$f(x)=\begin{vmatrix}\sin^2x & 1+\cos^2x & \cos 2x\\ 1+\sin^2x & \cos^2x & \cos 2x\\ \sin^2x & \cos^2x & \sin 2x\end{vmatrix},\ x\in R \text{ is :}$$

Open solution

(A) $\sqrt{5}$ (B) $\frac{3}{4}$ (C) 5 (D) $\sqrt{7}$

Q41. Numerical

Let $\frac{1}{16}$, a and b be in G.P. and $\frac{1}{a}$, $\frac{1}{b}$, 6 be in A.P., where $a, b > 0$. Then $72(a + b)$ is equal to _____.

Open solution

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Q42. Numerical

Open solution

For real numbers α, β, γ and δ, if

$$\int \frac{(x^2-1)+\tan^{-1}\left(\frac{x^2+1}{x}\right)}{(x^4+3x^2+1)\tan^{-1}\left(\frac{x^2+1}{x}\right)}dx$$

$$= \alpha\log_e\left(\tan^{-1}\left(\frac{x^2+1}{x}\right)\right) + \beta\tan^{-1}\left(\frac{\gamma(x^2-1)}{x}\right) + \delta\tan^{-1}\left(\frac{x^2+1}{x}\right) + C$$

where C is an arbitrary constant, then the value of $10(\alpha + \beta\gamma + \delta)$ is equal to _____.

Q43. Numerical

Open solution

Let f : R $\rightarrow$ R and g : R $\rightarrow$ R be defined as

$$f(x) = \begin{cases} x + a, & x < 0 \\ |x - 1|, & x \geq 0 \end{cases} \text{ and}$$

$$g(x) = \begin{cases} x + 1, & x < 0 \\ (x-1)^2 + b, & x \geq 0 \end{cases},$$

where a, b are non-negative real numbers. If (gof) (x) is continuous for all x $\in$ R, then a + b is equal to _____.

Q44. Numerical

Open solution

Consider the statistics of two sets of observations as follows :

	Size	Mean	Variance
Observation I	10	2	2
Observation II	n	3	1

If the variance of the combined set of these two observations is $\frac{17}{9}$, then the value of n is equal to _____.

Q45. Numerical

Open solution

Let $A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ be two 2 $\times$ 1 matrices with real entries such that A = XB, where

$X = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$, and k$\in$R.

If $a_1^2 + a_2^2 = \frac{2}{3}(b_1^2 + b_2^2)$ and $(k^2 + 1)\, b_2^2 \neq -2b_1b_2$, then the value of k is _____.

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Physics 25 questions

Q46. MCQ (Single Correct Answer)

What will be the nature of flow of water from a circular tap, when its flow rate increased from 0.18 L/min to 0.48 L/min? The radius of the tap and viscosity of water are 0.5 cm and 10^{-3} Pa s, respectively. (Density of water : 10^3 kg/m^3)

Open solution

(A) Steady flow to unsteady flow
(B) Unsteady to steady flow
(C) Remains turbulent flow
(D) Remains steady flow

Q47. MCQ (Single Correct Answer)

The refractive index of a converging lens is 1.4. What will be the focal length of this lens if it is placed in a medium of same refractive index? Assume the radii of curvature of the faces of lens are R_1 and R_2 respectively.

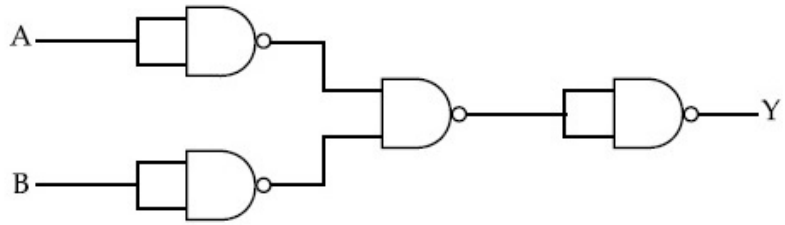
Open solution

(A) Zero
(B) Infinite
(C) 1
(D) $\frac{R_1R_2}{R_1-R_2}$

Q48. MCQ (Single Correct Answer)

The following logic gate is equivalent to :

Open solution



(A) NOR Gate
(B) NAND Gate
(C) OR Gate
(D) AND Gate

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Q49. MCQ (Single Correct Answer)

Open solution

Statement I : A cyclist is moving on an unbanked road with a speed of 7 kmh^{-1} and takes a sharp circular turn along a path of radius of 2m without reducing the speed. The static friction coefficient is 0.2. The cyclist will not slip and pass the curve. ($g = 9.8\ m/s^2$)

Statement II : If the road is banked at an angle of 45°, cyclist can cross the curve of 2m radius with the speed of 18.5 kmh^{-1} without slipping.

In the light of the above statements, choose the correct answer from the options given below.

(A) Statement I is incorrect and statement II is correct

(B) Both statement I and statement II are true

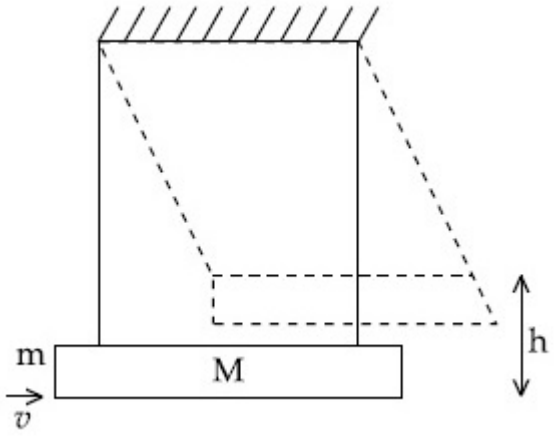
(C) Statement I is correct and statement II is incorrect

(D) Both statement I and statement II are false

Q50. MCQ (Single Correct Answer)

Open solution

A large block of wood of mass M = 5.99 kg is hanging from two long massless cords. A bullet of mass m = 10 g is fired into the block and gets embedded in it. The (block + bullet) then swing upwards, their centre of mass rising a vertical distance h = 9.8 cm before the (block + bullet) pendulum comes momentarily to rest at the end of its arc. The speed of the bullet just before collision is :
(take g = 9.8 ms^{-2})



(A) 831.4 m/s (B) 811.4 m/s (C) 841.4 m/s (D) 821.4 m/s

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Q51. MCQ (Single Correct Answer)

A mosquito is moving with a velocity $\vec{v} = 0.5t^2\hat{i} + 3t\hat{j} + 9\hat{k}$ m/s and accelerating in uniform conditions. What will be the direction of mosquito after 2 s?

(A) $\tan^{-1}\left(\frac{\sqrt{85}}{6}\right)$ from y-axis

(B) $\tan^{-1}\left(\frac{5}{2}\right)$ from y-axis

(C) $\tan^{-1}\left(\frac{2}{3}\right)$ from x-axis

(D) $\tan^{-1}\left(\frac{5}{2}\right)$ from x-axis

Open solution

Q52. MCQ (Single Correct Answer)

Red light differs from blue light as they have :

(A) Different frequencies and same wavelengths

(B) Different frequencies and different wavelengths

(C) Same frequencies and different wavelengths

(D) Same frequencies and same wavelengths

Open solution

Q53. MCQ (Single Correct Answer)

The de-Broglie wavelength associated with an electron and a proton were calculated by accelerating them through same potential of 100 V. What should nearly be the ratio of their wavelengths? (m_p = 1.00727u m_e = 0.00055u)

(A) 41.4 : 1

(B) $(1860)^2$: 1

(C) 1860 : 1

(D) 43 : 1

Open solution

Q54. MCQ (Single Correct Answer)

Calculate the value of mean free path (λ) for oxygen molecules at temperature 27°C and pressure 1.01×10^5 Pa. Assume the molecular diameter 0.3 nm and the gas is ideal. (k = 1.38×10^{-23} JK $^{-1}$)

(A) 32 nm

(B) 58 nm

(C) 86 nm

(D) 102 nm

Open solution

Q55. MCQ (Single Correct Answer)

A charge Q is moving $\vec{dl}$ distance in the magnetic field $\vec{B}$. Find the value of work done by $\vec{B}$.

(A) Zero

(B) -1

(C) Infinite

(D) 1

Open solution

Q56. MCQ (Single Correct Answer)

In order to determine the Young's Modulus of a wire of radius 0.2 cm (measured using a scale of least count = 0.001 cm) and length 1m (measured using a scale of least count = 1 mm), a weight of mass 1 kg (measured using a scale of least count = 1 g) was hanged to get the elongation of 0.5 cm (measured using a scale of least count 0.001 cm). What will be the fractional error in the value of Young's Modulus determined by this experiment?

(A) 0.14%

(B) 9%

(C) 1.4%

(D) 0.9%

Open solution

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Q57. MCQ (Single Correct Answer)

A resistor develops 500 J of thermal energy in 20 s when a current of 1.5A is passed through it. If the current is increased from 1.5A to 3A, what will be the energy developed in 20 s.

(A) 1000 J (B) 2000 J (C) 1500 J (D) 500 J

Open solution

Q58. MCQ (Single Correct Answer)

Find out the surface charge density at the intersection of point x = 3 m plane and x-axis, in the region of uniform line charge of 8 nC/m lying along the z-axis in free space.

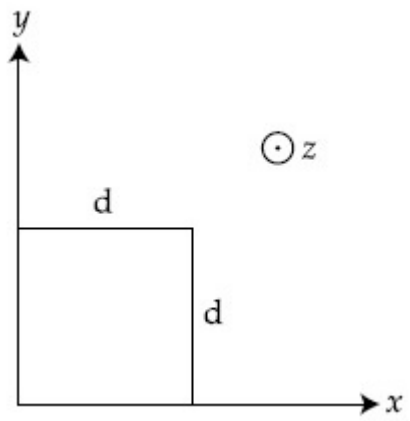
(A) 0.424 nC m^{-2} (B) 4.0 nC m^{-2} (C) 47.88 C/m (D) 0.07 nC m^{-2}

Open solution

Q59. MCQ (Single Correct Answer)

The magnetic field in a region is given by $\overrightarrow{B} = B_o\left(\frac{x}{a}\right)\hat{k}$. A square loop of side d is placed with its edges along the x and y axes. The loop is moved with a constant velocity $\overrightarrow{v} = v_0\hat{i}$. The emf induced in the loop is :

Open solution

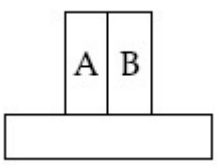


(A) $\frac{B_o v_o d^2}{2a}$ (B) $\frac{B_o v_o^2 d}{2a}$ (C) $\frac{B_o v_o d}{2a}$ (D) $\frac{B_o v_o d^2}{a}$

Q60. MCQ (Single Correct Answer)

A bimetallic strip consists of metals A and B. It is mounted rigidly as shown. The metal A has higher coefficient of expansion compared to that of metal B. When the bimetallic strip is placed in a cold bath, it will :

Open solution



(A) Neither bend nor shrink (B) Bend towards the left

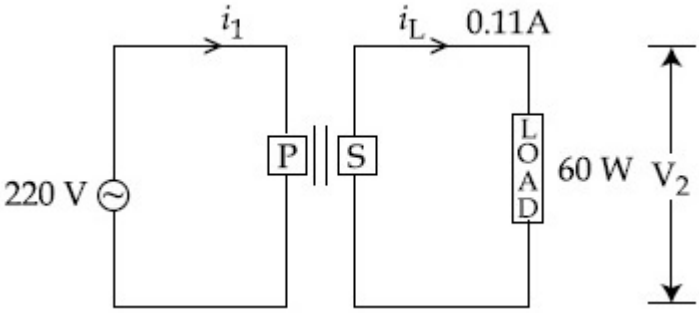
(C) Not bend but shrink (D) Bend towards the right

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Q61. MCQ (Single Correct Answer)

For the given circuit, comment on the type of transformer used.

Open solution



(A) Auxiliary transformer

(B) Step down transformer

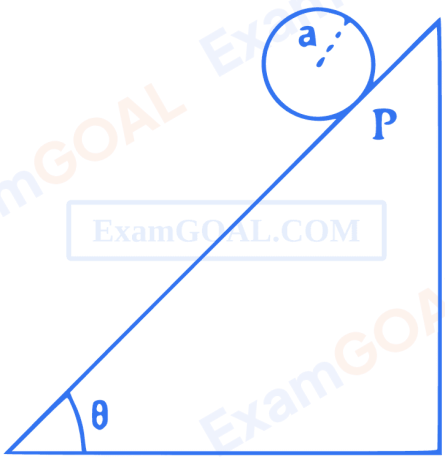
(C) Step-up transformer

(D) Auto transformer

Q62. Numerical

A solid disc of radius 'a' and mass 'm' rolls down without slipping on an inclined plane making an angle θ with the horizontal. The acceleration of the disc will be $\frac{2}{b}$g sinθ where b is _____.
(Round off to the Nearest Integer) (g = acceleration due to gravity, θ = angle as shown in figure)

Open solution



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Q63. Numerical

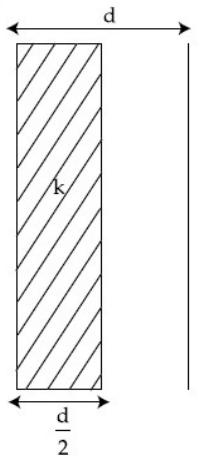
The energy dissipated by a resistor is 10 mJ in 1 s when an electric current of 2 mA flows through it. The resistance is _____Ω. (Round off to the Nearest Integer)

Open solution

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Q64. Numerical

In a parallel plate capacitor set up, the plate area of capacitor is 2 m^2 and the plates are separated by 1 m. If the space between the plates are filled with a dielectric material of thickness 0.5 m and area 2 m^2 (see fig.) the capacitance of the set-up will be _____ε_0. (Dielectric constant of the material = 3.2) (Round off to the Nearest Integer)



Open solution

Q65. Numerical

If one wants to remove all the mass of the earth to infinity in order to break it up completely.

The amount of energy that needs to be supplied will be $\frac{x}{5}\frac{GM^2}{R}$ where x is _____ (Round off to the Nearest Integer) (M is the mass of earth, R is the radius of earth, G is the gravitational constant)

Open solution

Q66. Numerical

A body of mass 2 kg moves under a force of $\left(2\hat{i}+3\hat{j}+5\hat{k}\right)$N. It starts from rest and was at the origin initially. After 4s, its new coordinates are (8, b, 20). The value of b is _____. (Round off to the Nearest Integer)

Open solution

Q67. Numerical

A closed organ pipe of length L and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases are equal in both the pipes. Both the pipes are vibrating in their first overtone with same frequency. The length of the open pipe is $\frac{x}{3}L\sqrt{\frac{\rho_1}{\rho_2}}$ where x is _____. (Round off to the Nearest Integer)

Open solution

Q68. Numerical

A swimmer can swim with velocity of 12 km/h in still water. Water flowing in a river has velocity 6 km/h. The direction with respect to the direction of flow of river water he should swim in order to reach the point on the other bank just opposite to his starting point is _____°. (Round off to the Nearest Integer) (Find the angle in degrees)

Open solution

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Q69. Numerical

A deviation of 2° is produced in the yellow ray when prism of crown and flint glass are achromatically combined. Taking dispersive powers of crown and flint glass as 0.02 and 0.03 respectively and refractive index for yellow light for these glasses are 1.5 and 1.6 respectively. The refracting angles for crown glass prism will be _____$^\circ$ (in degree). (Round off to the Nearest Integer)

Open solution

Q70. Numerical

A force $\overrightarrow{F} = 4\hat{i} + 3\hat{j} + 4\hat{k}$ is applied on an intersection point of x = 2 plane and x-axis. The magnitude of torque of this force about a point (2, 3, 4) is _____. (Round off to the Nearest Integer)

Open solution

Out of syllabus 20 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q71. MCQ (Single Correct Answer) · **Chemistry**

The INCORRECT statement regarding the structure of C_{60} is :

(A) The five-membered rings are fused only to six-membered rings.

(B) It contains 12 six-membered rings and 24 five-membered rings.

(C) Each carbon atom forms three sigma bonds.

(D) The six-membered rings are fused to both six and five-membered rings.

Open solution

Q72. MCQ (Single Correct Answer) · **Chemistry**

The green house gas/es is (are) :

(A) Carbon dioxide

(B) Oxygen

(C) Water vapour

(D) Methane

Choose the most appropriate answer from the options given below :

(A) (A) and (C) only

(B) (A) and (B) only

(C) (A), (C) and (D) only

(D) (A) only

Open solution

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Q73. MCQ (Single Correct Answer) · **Chemistry**

The correct statements about H_2O_2 are :

(A) used in the treatment of effluents.

(B) used as both oxidising and reducing agents.

(C) the two hydroxyl groups lie in the same plane.

(D) miscible with water.

Choose the correct answer from the options given below :

(A) (A), (C) and (D) only
(B) (A), (B), (C) and (D)
(C) (A), (B) and (D) only
(D) (B), (C) and (D) only

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

Which of the following polymer is used in the manufacture of wood laminates?

(A) Melamine formaldehyde resin
(B) Urea formaldehyde resin
(C) cis-poly isoprene
(D) Phenol and formaldehyde resin

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

The INCORRECT statements below regarding colloidal solutions is :

(A) The flocculating power of Al^{3+} is more than that of Na^{+}.
(B) A colloidal solutions shows Brownian motion of colloidal particles.
(C) An ordinary filter paper can stop the flow of colloidal particles.
(D) A colloidal solutions shows colligative properties.

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Which of the following reduction reaction CANNOT be carried out with coke?

(A) $ZnO \rightarrow Zn$
(B) $Fe_2O_3 \rightarrow Fe$
(C) $Cu_2O \rightarrow Cu$
(D) $Al_2O_3 \rightarrow Al$

Open solution

Q77. Numerical · **Chemistry**

Ga (atomic mass 70 u) crystallizes in a hexagonal close packed structure. The total number of voids in 0.581 g of Ga is _____ $\times 10^{21}$. (Round off to the Nearest Integer). [Given : $N_A = 6.023 \times 10_{23}$]

Open solution

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Q78. MCQ (Single Correct Answer) · **Mathematics**

If the points of intersections of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = 4b$, $b > 4$ lie on the curve $y^2 = 3x^2$, then b is equal to :

(A) 12 (B) 10 (C) 6 (D) 5

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

If (x, y, z) be an arbitrary point lying on a plane P which passes through the points (42, 0, 0), (0, 42, 0) and (0, 0, 42), then the value of the expression $3 + \frac{x-11}{(y-19)^2(z-12)^2} + \frac{y-19}{(x-11)^2(z-12)^2} + \frac{z-12}{(x-11)^2(y-19)^2} - \frac{x+y+z}{14(x-11)(y-19)(z-12)}$ is equal to :

(A) 3 (B) 39 (C) -45 (D) 0

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

Let C be the locus of the mirror image of a point on the parabola $y^2 = 4x$ with respect to the line $y = x$. Then the equation of tangent to C at P(2, 1) is :

(A) $x - y = 1$ (B) $2x + y = 5$ (C) $x + 3y = 5$ (D) $x + 2y = 4$

Open solution

Q81. Numerical · **Mathematics**

$S_n(x) = \log_{a^{1/2}}x + \log_{a^{1/3}}x + \log_{a^{1/6}}x + \log_{a^{1/11}}x + \log_{a^{1/18}}x + \log_{a^{1/27}}x +$ ..... up to n-terms, where $a > 1$. If $S_{24}(x) = 1093$ and $S_{12}(2x) = 265$, then value of a is equal to _____.

Open solution

Q82. Numerical · **Mathematics**

If the distance of the point $(1, -2, 3)$ from the plane $x + 2y - 3z + 10 = 0$ measured parallel to the line, $\frac{x-1}{3} = \frac{2-y}{m} = \frac{z+3}{1}$ is $\sqrt{\frac{7}{2}}$, then the value of $|m|$ is equal to _____.

Open solution

Q83. Numerical · **Mathematics**

Let $\vec{c}$ be a vector perpendicular to the vectors, $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. If $\vec{c} \cdot \left(\hat{i} + \hat{j} + 3\hat{k}\right) = 8$ then the value of $\vec{c} \cdot \left(\vec{a} \times \vec{b}\right)$ is equal to _____.

Open solution

Q84. Numerical · **Mathematics**

In ΔABC, the lengths of sides AC and AB are 12 cm and 5 cm, respectively. If the area of ΔABC is 30 cm^2 and R and r are respectively the radii of circumcircle and incircle of ΔABC, then the value of $2R + r$ (in cm) is equal to _____.

Open solution

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Q85. Numerical · **Mathematics**

Let n be a positive integer. Let

$A = \sum_{k=0}^{n} (-1)^k {}^nC_k \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k \right]$. If

$63A = 1 - \frac{1}{2^{30}}$, then n is equal to _____.

Open solution

Q86. MCQ (Single Correct Answer) · **Physics**

Two identical antennas mounted on identical towers are separated from each other by a distance of 45 km. What should nearly be the minimum height of receiving antenna to receive the signals in line of sight? (Assume radius of earth is 6400 km)

(A) 158.2 m (B) 79.1 m (C) 19.77 m (D) 39.55 m

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

Calculate the time interval between 33% decay and 67% decay if half-life of a substance is 20 minutes.

(A) 40 minutes (B) 60 minutes (C) 13 minutes (D) 20 minutes

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

The half-life of Au^{198} is 2.7 days. The activity of 1.50 mg of Au^{198} if its atomic weight is 198 g mol^{-1} is, ($N_a = 6 \times 10^{23}$/mol).

(A) 240 Ci (B) 357 Ci (C) 252 Ci (D) 535 Ci

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

Amplitude of a mass-spring system, which is executing simple harmonic motion decreases with time. If mass = 500g, Decay constant = 20 g/s then how much time is required for the amplitude of the system to drop to half of its initial value? (ln 2 = 0.693)

(A) 17.32 s (B) 34.65 s (C) 0.034 s (D) 15.01 s

Open solution

Q90. Numerical · **Physics**

For an ideal heat engine, the temperature of the source is 127°C. In order to have 60% efficiency the temperature of the sink should be _____°C. (Round off to the Nearest Integer)

Open solution

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Answer Key

Chemistry

1. **C**	2. **C**	3. **A**	4. **B**	5. **C**
6. **A**	7. **A**	8. **A**	9. **C**	10. **C**
11. **D**	12. **C**	13. **C**	14. **A**	15. **4**
16. **14**	17. **2218**	18. **3**	19. **108**	20. **19**
21. **525**	22. **1**	23. **19**		

Mathematics

24. **B**	25. **A**	26. **D**	27. **A**	28. **A**
29. **D**	30. **A**	31. **D**	32. **C**	33. **D**
34. **B**	35. **A**	36. **C**	37. **B**	38. **A**
39. **A**	40. **A**	41. **14**	42. **6**	43. **1**
44. **5**	45. **1**			

Physics

46. **A**	47. **B**	48. **A**	49. **B**	50. **A**
51. **A**	52. **B**	53. **D**	54. **D**	55. **A**
56. **C**	57. **B**	58. **A**	59. **D**	60. **B**
61. **C**	62. **3**	63. **2500**	64. **3**	65. **3**
66. **12**	67. **4**	68. **120**	69. **12**	70. **20**

Out of syllabus

71. **B**	72. **C**	73. **C**	74. **B**	75. **C**
76. **D**	77. **15**	78. **A**	79. **A**	80. **A**
81. **16**	82. **2**	83. **28**	84. **15**	85. **6**
86. **D**	87. **D**	88. **B**	89. **B**	90. **-113**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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